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Solution Of Body-Garment Collisions In Avatars For Immersive Reality Applications

Abstract

A method for resolving body-garment collisions in avatars for immersive reality applications is provided. The method includes forming a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset, identifying, from the two-dimensional projection, an area that includes a garment collision, and replacing a pixel in the area that includes the garment collision, with a pixel indicative of a garment for the dressed avatar, to form a new two-dimensional projection of the dressed avatar. A system and a non-transitory, computer-readable medium storing instructions to perform the above method, are also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/148,326, filed Dec. 29, 2022, titled “SOLUTION OF BODY-GARMENT COLLISIONS IN AVATARS FOR IMMERSIVE REALITY APPLICATIONS” currently pending, and which is herein incorporated by reference in its entirety.

BACKGROUND
Technical Field
[0002] The present disclosure is related to generating dressed avatars in virtual reality (VR) and augmented reality (AR) applications. More specifically, the present disclosure resolves body-garment collisions for VR/AR applications in real time.
Related Art
[0003] In the field of VR/AR real-time garment drapping, body-garment collisions are not infrequent. Resolving such occurrences is possible to a high degree of accuracy using three-dimensional meshes rendered for the garment and the subject body in the avatar. However, these approaches involve heavy computation that are currently not feasible in real time.
SUMMARY
[0004] In a first embodiment, a computer-implemented method includes forming a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset, identifying, from the two-dimensional projection, an area that includes a garment collision, and replacing a pixel in the area that includes the garment collision, with a pixel indicative of a garment for the dressed avatar, to form a new two-dimensional projection of the dressed avatar.
[0005] In a second embodiment, a system includes a memory storing multiple instructions and one or more processors configured to execute the instructions to cause the system to perform operations. The operations include to form a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset, to identify, from the two-dimensional projection, an area that includes a garment collision, and to replace a pixel in the area that includes the garment collision, with a pixel indicative of a garment for the dressed avatar, to form a new two-dimensional projection of the dressed avatar.
[0006] In a third embodiment, a computer-implemented method for training a model to remove garment collisions in a virtual reality headset includes collecting multiple images of a dressed avatar, each of the images having at least one known garment collision, identifying, with the model, a selected area of each of the images that contains the garment collision, modifying a first feature in the model when at least one pixel of the at least one known garment collision is not included in the selected area, and modifying a second feature in the model when at least one pixel in the selected area is not part of at least one garment collision.
[0007] In another embodiment, a non-transitory, computer-readable medium stores instructions which, when executed by a processor in a computer, cause the computer to perform a method. The method includes forming a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset, identifying, from the two-dimensional projection, an area that includes a garment collision, generating a mask for multiple pixels in the area that includes the garment collision, wherein at least one pixel in the mask has a value indicative of a lack of a garment portion, and replacing, in a new two-dimensional projection of the dressed avatar, a pixel having a binary value indicative of the lack of the garment portion with a pixel that is included in the garment portion.

[0008] In another embodiment, a system includes a first means to store instructions and a second means to execute the instructions to cause the system to perform a method. The method includes forming a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset, identifying, from the two-dimensional projection, an area that includes a garment collision, generating a mask for multiple pixels in the area that includes the garment collision, wherein at least one pixel in the mask has a value indicative of a lack of a garment portion, and replacing, in a new two-dimensional projection of the dressed avatar, a pixel having a binary value indicative of the lack of the garment portion with a pixel that is included in the garment portion.
[0009] In yet another embodiment, a system includes a first means to store instructions and a second means to execute the instructions to cause the system to perform a method. The method includes forming a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset, identifying, from the two-dimensional projection, an area that includes a garment collision, and replacing a pixel in the area that includes the garment collision with a pixel indicative of a garment for the dressed avatar to form a new two-dimensional projection of the dressed avatar.
[0010] These and other embodiments will be clear to one of ordinary skill in the art, in view of the following.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0011] FIG. **1** illustrates an example architecture suitable for providing a real-time, clothed subject animation in a virtual reality environment, according to some embodiments.
[0012] FIG. **2** is a block diagram illustrating an example server and client from the architecture of FIG. **1**, according to certain aspects of the disclosure.
[0013] FIG. **3** illustrates an architecture for a neural network tool to identify garment collisions, according to some embodiments.
[0014] FIGS. **4**A-**4**B illustrate a sequence for training a neural network for creating a mask over garment collisions in an avatar, according to some embodiments.
[0015] FIGS. **5**A-**5**B illustrate a sequence for cropping an area of interest in an image projection of a dressed avatar, according to some embodiments.
[0016] FIG. **6** illustrates a ray marching process for resolving a garment collision in a 3D domain, according to some embodiments.
[0017] FIG. **7** illustrates a resolution of a garment collision with a 2D model, according to some embodiments.
[0018] FIG. **8** illustrates a flowchart in a method for resolving body-garment collisions for avatars in an immersive application, according to some embodiments.
[0019] FIG. **9** illustrates a flowchart in a method for training a neural network to identify and resolve body-garment collisions for avatars in immersive applications, according to some embodiments.
[0020] FIG. **10** illustrates a computer system configured to perform at least some of the methods of FIGS. **8-9**, according to some embodiments.
[0021] In the figures, like elements are labeled likewise, according to their description, unless explicitly stated otherwise.
DETAILED DESCRIPTION OF THE FIGURES
[0022] In the following detailed description, numerous specific details are set forth to provide a full understanding of the present disclosure. It will be apparent, however, to one ordinarily skilled in the art, that embodiments of the present disclosure may be practiced without some of these specific

details. In other instances, well-known structures and techniques have not been shown in detail so as not to obscure the disclosure.

General Overview

[0023] In the metaverse, users express themselves by wearing different clothes on their avatars. For a successful immersive experience, realistic rendering of clothed avatars with high fidelity is desirable. However, the limited availability of computer power hinders this goal. Current machine learning (ML) clothing methods tend to wrongly dispose garments and body locations that intersect with each other at some points (e.g., "collide"), resulting in obvious, unrealistic errors. Typically, ML-clothing techniques train on ground-truth datasets of realistic body-meshes for physical simulations results. Accordingly, for each collision there exists a ground-truth correct configuration that can be incorporated into a loss function to resolve the collision problem.

[0024] Therefore, currently available ML methods apply lengthy post-processing steps to identify and correct garment and body mesh collisions. These steps are compute-intensive and inhibit the usage of such ML methods in real-time immersive applications. Accordingly, it is desirable to have computer-efficient approaches to solve clothing and body collisions for real-time metaverse personalized fashion experiences.

[0025] To solve the above technical problem arising in the field of immersive reality applications for computer networks, embodiments as disclosed herein implement an ML classification step to identify well defined areas of an image including a collision artifact by learning to classify bad ML predictions. Further, some embodiments include resolving the collision artifacts in the image-domain, rather than in the 3D mesh domain and training a module to fix the ML-clothing predictions. For example, ML-clothing methods predict the locations of the body and garment meshes, which are rendered on a two-dimensional (2D) headset display, for the user. Accordingly, collision artifacts can be resolved in a 2D space, for a single rendered image.

[0026] In some embodiments, a three-dimensional (3D) mesh collision problem is solved with a 2D supervised pixel classification ML model based on a given user viewpoint. The simplicity of the approach enables real-time rendering of avatars' clothed bodies, for an immersive user experience.

[0027] In some embodiments, a supervised per-pixel classification solution is provided in 2D. A training set including a body, a garment position, a camera position, and a created image is used in an ML-draping model to estimate a second body position and garment position to be used for animation of a dressed avatar in an immersive reality application. The second body position and the second garment position may include collision artifacts. The model is trained to identify artifacts in the second image based on the first image as a binary mask, indicating the pixels in the second image where the garment should replace the body. Accordingly, the model creates a new 2D image that no longer depicts collisions from the point of view of the user.

Example System Architecture

[0028] FIG. **1** illustrates an architecture **10** for immersive reality applications including a headset **100**, a mobile device **110**, a remote server **130** and a database **152**, according to some embodiments. Headset **100** may be a VR/AR headset running an immersive reality application, and mobile device **110** may be a smart phone with a user **101** of headset **100**. Headset **100**, mobile device **110**, server **130** and database **152** may communicate with one another via a communications module **118** and exchange a first dataset **103-1**. In some embodiments, communications module **118** can include, for example, radio-frequency hardware (e.g., antennas, filters, analog to digital converters, and the like) and software (e.g., signal processing software). Dataset **103-1** may include a video or a rendering of an immersive reality environment, or a view of a gesture, face, or gaze direction of user **101**. In some embodiments, user **101** is also the owner or is associated with mobile device **110**. In some embodiments, headset **100** may directly communicate with remote server **130**, database **152**, or any other client device (e.g., a smart phone of a different user, and the like) via a network **150**.

[0029] Mobile device **110** may be communicatively coupled with remote server **130** and database

152 via network **150**, and transmit/share information, files, and the like with one another (e.g., dataset **103-2** and dataset **103-3**). Datasets **103-1**, **103-2**, and **103-3** will be collectively referred to, hereinafter, as “datasets **103**.” In some embodiments, headset **100** may include multiple sensors **121** such as inertial measurement units (IMUs), gyroscopes, microphones, cameras, and the like mounted within the frame of headset **100**. Other sensors **121** that can be included in the wearable devices **100** may be magnetometers, photodiodes and cameras, touch sensors, and other electromagnetic devices such as capacitive sensors, a pressure sensor, and the like. Headset **100** may also include a microphone **125**. Accordingly, datasets **103** may include signals that locate a user **101** within the immersive reality application, including the user's point of view of an avatar.

[0030] In addition, headset **100**, and any other wearable device **100**, or mobile device **110** may include a memory circuit **120** storing instructions, and a processor circuit **112** configured to execute the instructions to cause headset **100** to perform, at least partially, some of the steps in methods consistent with the present disclosure. Headset **100** may be down-streaming a virtual reality environment from server **130** including dressed avatars from other participants in the immersive reality environment. Network **150** may include, for example, any one or more of a local area network (LAN), a wide area network (WAN), the Internet, and the like. Further, the network can include, but is not limited to, any one or more of the following network topologies, including a bus network, a star network, a ring network, a mesh network, a star-bus network, tree or hierarchical network, and the like.

[0031] In some embodiments, memory circuit **120** may store instructions which, when executed by processor **112**, cause the combination components to remove a garment collision for an avatar in the immersive reality application.

[0032] FIG. **2** is a block diagram **200** illustrating an example server **130** and client device **110** from architecture **100**, according to certain aspects of the disclosure. Client device **110** and server **130** are communicatively coupled over network **150** via respective communications modules **218-1** and **218-2** (hereinafter, collectively referred to as “communications modules **218**”). Communications modules **218** are configured to interface with network **150** to send and receive information, such as data, requests, responses, and commands to other devices via network **150**. Communications modules **218** can be, for example, modems or Ethernet cards, and may include radio hardware and software for wireless communications (e.g., via electromagnetic radiation, such as radiofrequency —RF—, near field communications —NFC—, Wi-Fi, and Bluetooth radio technology). A user may interact with client device **110** via an input device **214** and an output device **216**. Input device **214** may include a mouse, a keyboard, a pointer, a touchscreen, a microphone, a joystick, a virtual joystick, and the like. In some embodiments, input device **214** may include cameras, microphones, and sensors, such as touch sensors, acoustic sensors, inertial motion units —IMUs— and other sensors configured to provide input data to a VR/AR headset. For example, in some embodiments, input device **214** may include an eye tracking device to detect the position and gaze direction of a user's pupil in a VR/AR headset. Output device **216** may be a screen display, a touchscreen, a speaker, and the like. Client device **110** may include a memory **220-1** and a processor **212-1**. Memory **220-1** may include an application **222** and a GUI **225**, configured to run in client device **110** and couple with input device **214** and output device **216**. Application **222** may be downloaded by the user from server **130** and may be hosted by server **130**. In some embodiments, client device **110** is a VR/AR headset and application **222** is an immersive reality application.

[0033] Server **130** includes a memory **220-2**, a processor **212-2**, and communications module **218-2**. Hereinafter, processors **212-1** and **212-2**, and memories **220-1** and **220-2**, will be collectively referred to, respectively, as “processors **212**” and “memories **220**.” Processors **212** are configured to execute instructions stored in memories **220**. In some embodiments, memory **220-2** includes an avatar engine **232**. Avatar engine **232** may share or provide features and resources to GUI **225**, including multiple tools associated with training and using a three-dimensional avatar rendering model and a collision model **240** for immersive reality applications (e.g., application **222**). The user may access avatar engine **232** through application **222**, installed in a memory **220-1** of client device **110**. Accordingly, application **222**, including GUI **225**, may be installed by server **130** and perform scripts and other routines provided by server **130** through any one of multiple tools. Execution of application **222** may be controlled by processor **212-1**.

[0034] In that regard, avatar engine **232** may be configured to create, store, update, and maintain a collision model **240**, as disclosed herein. Collision model **240** may include an encoder-decoder tool **242**, a draping tool **244**, a mask tool **246**, and a neural network tool **248**. Encoder-decoder tool **242** collects input images with multiple, simultaneous views of a subject and extracts pixel-aligned features to condition draping tool **244** and identify garment collisions with mask tool **246**. Collision model **240** can generate novel views of unseen subjects from one or more sample images processed by encoder-decoder tool **242**. In some embodiments, encoder-decoder tool **242** is a shallow (e.g., including a few one-or two-node layers) convolutional network. In some embodiments, mask tool **246** identifies garment collisions and creates a mask that can be overlapped on a 2D projection of a subject avatar.

[0035] Neural network tool **248** may include algorithms trained for the specific purposes of the engines and tools included therein. The algorithms may include machine learning or artificial intelligence algorithms making use of any linear or non-linear algorithm, such as a neural network algorithm, or multivariate regression algorithm. In some embodiments, the machine learning model may include a neural network (NN), a convolutional neural network (CNN), a generative adversarial neural network (GAN), a deep reinforcement learning (DRL) algorithm, a deep recurrent neural network (DRNN), a classic machine learning algorithm such as random forest, k-nearest neighbor (KNN) algorithm, k-means clustering algorithms, or any combination thereof. More generally, the machine learning model may include any machine learning model involving a training step and an optimization step. In some embodiments, training database **252** may include a training archive to modify coefficients according to a desired outcome of the machine learning model. Accordingly, in some embodiments, neural network tool **248** is configured to access training database **252** to retrieve documents and archives as inputs for the machine learning model.

[0036] In some embodiments, avatar engine **232**, the tools contained therein, and at least part of training database **252** may be hosted in a different server that is accessible by server **130** or client device **110**. In some embodiments, avatar engine **232** may access one or more machine learning models stored in a training database **252**. Training database **252** includes training archives and other data files that may be used by avatar engine **232** in the training of a machine learning model, according to the input of the user through application **222**. Moreover, in some embodiments, at least one or more training archives or machine learning models may be stored in either one of memories **220** and the user may have access to them through application **222**.

[0037] FIG. **3** illustrates an architecture for a neural network **300** to identify garment collisions (cf. neural network tool **248**), according to some embodiments. A two-dimensional input image **301** is a projection along a given point of view of a dressed avatar **302** having garment collisions **311**, and NN **300** generates a mask **331** that includes pixels having binary values (e.g., 0 and 1), wherein a first value **321** indicates a body portion of avatar **302** wherein a garment portion should be included. A second value **322** indicates the absence of a garment collision.

[0038] NN **300** may include down-sampling convolutional steps **310** concatenated with up-sampling steps **320** in concatenation stages **312-1**, **312-2**, **312-3** and **312-4**.

[0039] FIGS. **4**A-**4**B illustrate a sequence for training a neural network **400** for creating a mask **431** over garment collisions **411** in an avatar **402**, according to some embodiments. An input image **401** along a selected point of view includes garment collisions **411**, wherein the body of avatar **402** overlaps a garment **444**. Neural network **400** may be a 2D to 2D network, and produces a mask **431**B to create an image output **431**A wherein garment collisions have been replaced by clothing **444**, as desired.

[0040] In some embodiments, NN **400** uses a physics simulation as ground-truth, to deduce mask output **431**. NN **400** is trained to classify pixels with collisions as “1,” and without collision as “0.” NN **400** takes, as input, multiple images **401** (e.g., taken from the output of an ML-draping algorithm) each paired with a mask **431**B that includes labeled pixels **421**B. The training pairs (**401**, **431**) may be generated by using multiple body shapes, multiple poses, and multiple viewing angles.

[0041] FIGS. **5**A-**5**B illustrate a sequence for cropping an area of interest **520** in an image projection **501** of a dressed avatar **502**, according to some embodiments. Area of interest **520** includes garment collision pixels **521**. Because the rest of the area in image **501** is collision-free, avatar engine **532**A crops the image, so the computational burden is substantially reduced. A cropped image **531**A is used as input in avatar engine **532**B (which may be the same, but a different processing block as avatar engine **532**A), to generate a collision free projection **531**. Finally, projection **531** is combined with the rest of the cropped projection **501** to provide a fully dressed (e.g., collision free) avatar image.

[0042] FIG. **6** illustrates a ray marching process **600** for resolving a garment collision in a 3D domain, according to some embodiments. An avatar model **602** includes collisions **611** between a body model **642** and a garment model **644**. Collisions **611** are **3**D objects having a volume. Assuming the physics of an immersive reality scene is known (specifically, the viewing direction of the user/camera, the intrinsic and extrinsic parameters, and the like), a point **650** may be selected as a point of view (e.g., camera's center, a user's pupil, and the like). Given a point **650**, each pixel on a 2D projection **630** of avatar **602** includes at most one ray **655** starting in point of view **650** and ending within avatar **602**. For each of the rays **655**, the method includes identifying whether the first contact with avatar **602** (when marching from point **650**, through projection **630**, and through avatar **602**) is a body portion or a garment portion. If ray **655**, proceeding in the above sequence includes a sequence of points: body-garment-body, then the first body-garment transition is identified as part of a garment-collision volume. The projection **620** of the garment-collision volume on projection **630** is then the 2D collision area to resolve, and includes a pixel **621** that masks collisions **611**.

[0043] FIG. **7** illustrates a resolution **700** of a garment collision with a 2D model **740**, according to some embodiments. 2D model **740** identifies collision areas **711** in a 2D image **701** of a dressed avatar **702**. Performing operations on the pixels within collision areas **711** (e.g., replacing, recoloring, or merging with the garment), 2D model **740** provides a fully dressed image **731** of avatar **702**.

[0044] FIG. **8** illustrates a flowchart in a method **800** for resolving body-garment collisions for avatars in an immersive application, according to some embodiments. In some embodiments, at least one step in method **800** may be performed by a processor executing instructions stored in a memory of a computer system in a VR/AR headset, a mobile device, a remote server or a database, communicatively coupled through a network via a communications module (cf. VR/AR headset **100**, mobile device **110**, server **130**, database **152**, processor **112**, memory **120**, communications module **118**, and network **150**). The memory may include instructions to run an immersive reality application and interact with the user via a GUI, or an avatar engine including a collision model, an encoder-decoder tool, a draping tool, a mask tool, and a neural network tool, as disclosed herein (cf. application **222**, GUI **225**, avatar engine **232**, collision model **240**, encoder-decoder tool **242**, draping tool **244**, mask tool **246**, and neural network tool **248**). Methods consistent with the present disclosure may include at least one or more of the steps in method **800** performed in a different order, simultaneously, quasi-simultaneously, or overlapping in time.

[0045] Step **802** includes forming a two-dimensional projection of a dressed avatar in an immersive reality application running in a headset. In some embodiments, step **802** includes selecting the two-dimensional projection along a direction of view of a user of the headset, based on the immersive reality application.

[0046] Step **804** includes identifying, from the two-dimensional projection, an area that includes a garment collision. In some embodiments, step **804** includes running a two-dimensional model trained to identify a garment collision on the two-dimensional projection of the dressed avatar. In some embodiments, step **804** includes cropping one or more portions of the two-dimensional projection, each of the one or more portions including multiple adjacent pixels with a value indicative of a lack of the garment portion. In some embodiments, step **804** includes cropping the two-dimensional projection around the area that includes the garment collision to form a reduced image input, and applying a two-dimensional model to the reduced image input for replacing the first pixel having the binary value indicative of the lack of the garment portion. In some embodiments, step **804** includes generating a mask for multiple pixels in the area that includes the garment collision, wherein at least one pixel in the mask has a value indicative of a lack of a garment portion. In some embodiments, step **806** includes predicting, for each pixel, a likelihood that the pixel indicates a body portion, based on a pixel value and at least an adjacent pixel value.

[0047] Step **808** includes replacing a pixel in the area that includes the garment collision with a pixel indicative of a garment for the dressed avatar, to form a new two-dimensional projection of the dressed avatar. In some embodiments, step **808** includes determining a color value and a shade value for the first pixel, based on a two-dimensional garment model that includes the garment portion. In some embodiments, step **808** includes resolving an overlap between a three-dimensional garment model and a three-dimensional body model, and projecting an updated three-dimensional garment model on the new two-dimensional projection of the dressed avatar. In some embodiments, step **808** includes replacing the garment portion using a two-dimensional garment model. In some embodiments, step **808** includes coloring the first pixel according to an average color of a garment in an area adjacent to the area of the garment collision.

[0048] FIG. **9** illustrates a flowchart in a method **900** for training a neural network to identify and resolve body-garment collisions for avatars in immersive applications, according to some embodiments. In some embodiments, at least one step in method **900** may be performed by a processor executing instructions stored in a memory of a computer system in a VR/AR headset, a mobile device, a remote server or a database, communicatively coupled through a network via a communications module (cf. VR/AR headset **100**, mobile device **110**, server **130**, database **152**, processor **112**, memory **120**, communications module **118**, and network **150**). The memory may include instructions to run an immersive reality application and interact with the user via a GUI, or an avatar engine including a collision model, an encoder-decoder tool, a draping tool, a mask tool, and a neural network tool, as disclosed herein (cf. application **222**, GUI **225**, avatar engine **232**, collision model **240**, encoder-decoder tool **242**, draping tool **244**, mask tool **246**, and neural network tool **248**). Methods consistent with the present disclosure may include at least one or more of the steps in method **900** performed in a different order, simultaneously, quasi-simultaneously, or overlapping in time.

[0049] Step **902** includes collecting multiple images of a dressed avatar, each of the images having at least one known garment collision. In some embodiments, step **902** includes collecting images with different body shapes, different poses, and different viewing angles. In some embodiments, step **902** includes projecting the dressed avatar on a two-dimensional plane along a point of view of an observer in an immersive reality application. In some embodiments, step **902** includes generating a three-dimensional model for a garment of an avatar body that overlaps the avatar body in a finite volume.

[0050] Step **904** includes identifying, with the model, a selected area of each of the images that contains the garment collision, wherein the model includes a loss function including the images of the dressed avatar as ground truth data. In some embodiments, the ground-truth data used in the model may include physics simulations of a dressed avatar.

[0051] Step **906** includes identifying a first coefficient in the model when at least one pixel of the at least one known garment collision is not included in the selected area.

[0052] Step **908** includes modifying a second coefficient in the model when at least one pixel in the selected area is not part of at least one garment collision. In some embodiments, step **908** includes defining a loss function for each pixel in the selected area. In some embodiments, step **908** includes projecting a three-dimensional garment model to form a two-dimensional garment view along a selected point of view to update the at least one pixel in the selected area based on the two-dimensional garment view. In some embodiments, step **908** includes verifying that the at least one pixel of the at least one known garment collision is not included in the selected area when a selected point of view of the dressed avatar is modified.

Hardware Overview

[0053] FIG. **10** is a block diagram illustrating an exemplary computer system **1000** with which headsets and other client devices **110**, and methods **800** and **900** can be implemented. In certain aspects, computer system **1000** may be implemented using hardware or a combination of software and hardware, either in a dedicated server, or integrated into another entity, or distributed across multiple entities. Computer system **1000** may include a desktop computer, a laptop computer, a tablet, a phablet, a smartphone, a feature phone, a server computer, or otherwise. A server computer may be located remotely in a data center or be stored locally.

[0054] Computer system **1000** includes a bus **1008** or other communication mechanism for communicating information, and a processor **1002** (e.g., processors **212**) coupled with bus **1008** for processing information. By way of example, the computer system **1000** may be implemented with one or more processors **1002**. Processor **1002** may be a general- purpose microprocessor, a microcontroller, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Programmable Logic Device (PLD), a controller, a state machine, gated logic, discrete hardware components, or any other suitable entity that can perform calculations or other manipulations of information.

[0055] Computer system **1000** can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them stored in an included memory **1004** (e.g., memories **220**), such as a Random Access Memory (RAM), a flash memory, a Read-Only Memory (ROM), a Programmable Read-Only Memory (PROM), an Erasable PROM (EPROM), registers, a hard disk, a removable disk, a CD-ROM, a DVD, or any other suitable storage device, coupled with bus **1008** for storing information and instructions to be executed by processor **1002**. The processor **1002** and the memory **1004** can be supplemented by, or incorporated in, special purpose logic circuitry.

[0056] The instructions may be stored in the memory **1004** and implemented in one or more computer program products, e.g., one or more modules of computer program instructions encoded on a computer-readable medium for execution by, or to control the operation of, the computer system **1000**, and according to any method well known to those of skill in the art, including, but not limited to, computer languages such as data-oriented languages (e.g., SQL, dBase), system languages (e.g., C, Objective-C, C++, Assembly), architectural languages (e.g., Java, .NET), and application languages (e.g., PHP, Ruby, Perl, Python). Instructions may also be implemented in computer languages such as array languages, aspect-oriented languages, assembly languages, authoring languages, command line interface languages, compiled languages, concurrent languages, curly-bracket languages, dataflow languages, data-structured languages, declarative languages, esoteric languages, extension languages, fourth-generation languages, functional languages, interactive mode languages, interpreted languages, iterative languages, list-based languages, little languages, logic-based languages, machine languages, macro languages, metaprogramming languages, multiparadigm languages, numerical analysis, non-English-based languages, object-oriented class-based languages, object-oriented prototype-based languages, off-side rule languages, procedural languages, reflective languages, rule-based languages, scripting languages, stack-based languages, synchronous languages, syntax handling languages, visual languages, wirth languages, and xml-based languages. Memory **1004** may also be used for storing temporary variable or other intermediate information during execution of instructions to be executed by processor **1002**.

[0057] A computer program as discussed herein does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, subprograms, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network. The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output.

[0058] Computer system **1000** further includes a data storage device **1006** such as a magnetic disk or optical disk, coupled with bus **1008** for storing information and instructions. Computer system **1000** may be coupled via input/output module **1010** to various devices. Input/output module **1010** can be any input/output module. Exemplary input/output modules **1010** include data ports such as USB ports. The input/output module **1010** is configured to connect to a communications module **1012**. Exemplary communications modules **1012** include networking interface cards, such as Ethernet cards and modems. In certain aspects, input/output module **1010** is configured to connect to a plurality of devices, such as an input device **1014** and/or an output device **1016**. Exemplary input devices **1014** include a keyboard and a pointing device, e.g., a mouse or a trackball, by which a consumer can provide input to the computer system **1000**. Other kinds of input devices **1014** can be used to provide for interaction with a consumer as well, such as a tactile input device, visual input device, audio input device, or brain-computer interface device. For example, feedback provided to the consumer can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the consumer can be received in any form, including acoustic, speech, tactile, or brain wave input. Exemplary output devices **1016** include display devices, such as an LCD (liquid crystal display) monitor, for displaying information to the consumer.

[0059] According to one aspect of the present disclosure, headsets and client devices **110** can be implemented, at least partially, using a computer system **1000** in response to processor **1002** executing one or more sequences of one or more instructions contained in memory **1004**. Such instructions may be read into memory **1004** from another machine-readable medium, such as data storage device **1006**. Execution of the sequences of instructions contained in main memory **1004** causes processor **1002** to perform the process steps described herein. One or more processors in a multi-processing arrangement may also be employed to execute the sequences of instructions contained in memory **1004**. In alternative aspects, hard-wired circuitry may be used in place of or in combination with software instructions to implement various aspects of the present disclosure. Thus, aspects of the present disclosure are not limited to any specific combination of hardware circuitry and software.

[0060] Various aspects of the subject matter described in this specification can be implemented in a computing system that includes a back end component, e.g., a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical consumer interface or a Web browser through which a consumer can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. The communication network can include, for example, any one or more of a LAN, a WAN, the Internet, and the like. Further, the communication network can include, but is not limited to, for example, any one or more of the following network topologies, including a bus network, a star network, a ring network, a mesh network, a star-bus network, tree or hierarchical network, or the like. The communications modules can be, for example, modems or Ethernet cards.

[0061] Computer system **1000** can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. Computer system **1000** can be, for example, and without limitation, a desktop computer, laptop computer, or tablet computer. Computer system **1000** can also be embedded in another device, for example, and without limitation, a mobile telephone, a PDA, a mobile audio player, a Global Positioning System (GPS) receiver, a video game console, and/or a television set top box.

[0062] The term "machine-readable storage medium" or "computer-readable medium" as used herein refers to any medium or media that participates in providing instructions to processor **1002** for execution. Such a medium may take many forms, including, but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical or magnetic disks, such as data storage device **1006**. Volatile media include dynamic memory, such as memory **1004**. Transmission media include coaxial cables, copper wire, and fiber optics, including the wires forming bus **1008**. Common forms of machine-readable media include, for example, floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH EPROM, any other memory chip or cartridge, or any other medium from which a computer can read. The machine-readable storage medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter affecting a machine-readable propagated signal, or a combination of one or more of them.

[0063] To illustrate the interchangeability of hardware and software, items such as the various illustrative blocks, modules, components, methods, operations, instructions, and algorithms have been described generally in terms of their functionality. Whether such functionality is implemented as hardware, software, or a combination of hardware and software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application.

[0064] As used herein, the phrase "at least one of" preceding a series of items, with the terms "and" or "or" to separate any of the items, modifies the list as a whole, rather than each member of the list (e.g., each item). The phrase "at least one of" does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases "at least one of A, B, and C" or "at least one of A, B, or C" each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0065] The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any embodiment described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments. Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, and other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0066] A reference to an element in the singular is not intended to mean “one and only one” unless specifically stated, but rather “one or more.” The term “some” refers to one or more. Underlined and/or italicized headings and subheadings are used for convenience only, do not limit the subject technology, and are not referred to in connection with the interpretation of the description of the subject technology. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. All structural and functional equivalents to the elements of the various configurations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the subject technology. Moreover, nothing disclosed herein is intended to be dedicated to the public, regardless of whether such disclosure is explicitly recited in the above description. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

[0067] While this specification contains many specifics, these should not be construed as limitations on the scope of what may be described, but rather as descriptions of particular implementations of the subject matter. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially described as such, one or more features from a described combination can in some cases be excised from the combination, and the described combination may be directed to a subcombination or variation of a subcombination.

[0068] The subject matter of this specification has been described in terms of particular aspects, but other aspects can be implemented and are within the scope of the following claims. For example, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. The actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the aspects described above should not be understood as requiring such separation in all aspects, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0069] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the described subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately described subject matter.

[0070] The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims and to encompass all legal equivalents.

Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

Claims

1-20. (canceled)

21. A computer-readable storage medium storing instructions, for training a model to remove garment collisions in a virtual reality headset, the instructions, when executed by a computing system, cause the computing system to: collect multiple images of a dressed avatar, each of the images having at least one known garment collision; and for each particular image of the multiple images: identify, with the model, a selected area of the particular image that the model estimates contains a garment collision, wherein the model utilizes the images of the dressed avatar with the known garment collision as ground-truth data by: modifying a first feature in the model when a first portion of the selected area of particular image is within the known garment collision for the particular image; and modifying second first feature in the model when a second portion of the selected area of particular image is not within the known garment collision for the particular image; wherein modifying the first feature and the second feature includes defining a loss function based on the first portion or second portion.

22. The computer-readable storage medium of claim 21, wherein the collecting multiple images of the dressed avatar includes collecting images with different body shapes, different poses, and different viewing angles.

23. The computer-readable storage medium of claim 21, wherein the collecting multiple images of the dressed avatar includes projecting the dressed avatar on a two-dimensional plane along a point of view of an observer in an immersive reality application.

24. The computer-readable storage medium of claim 21, wherein the collecting multiple images of the dressed avatar includes generating a three-dimensional model for a garment of an avatar body that overlaps the avatar body in a finite volume.

25. The computer-readable storage medium of claim 21, wherein the modifying the second feature in the model includes projecting a three- dimensional garment model to form a two-dimensional garment view along a selected point of view to update the at least one pixel in the selected area based on the two-dimensional garment view.

26. The computer-readable storage medium of claim 21, wherein the modifying the first feature in the model includes verifying that at least one pixel of the at least one known garment collision is not included in the selected area when a selected point of view of the dressed avatar is modified.

27. A computing system for training a model to remove garment collisions in a virtual reality headset, the computing system comprising: one or more processors; and one or more memories storing instructions that, when executed by the one or more processors, cause the computing system to: collect multiple images of a dressed avatar, each of the images having at least one known garment collision; and for each particular image of the multiple images: identify, with the model, a selected area of the particular image that the model estimates contains a garment collision, wherein the model utilizes the images of the dressed avatar with the known garment collision as ground-truth data by: modifying a first feature in the model when a first portion of the selected area of particular image is within the known garment collision for the particular image; and modifying second first feature in the model when a second portion of the selected area of particular image is not within the known garment collision for the particular image; wherein modifying the first feature and the second feature includes defining a loss function based on the first portion or second portion.

28. The computing system of claim 27, wherein the collecting multiple images of the dressed avatar includes collecting images with different body shapes, different poses, and different viewing angles.

29. The computing system of claim 27, wherein the collecting multiple images of the dressed avatar includes projecting the dressed avatar on a two-dimensional plane along a point of view of an observer in an immersive reality application.
30. The computing system of claim 27, wherein the collecting multiple images of the dressed avatar includes generating a three-dimensional model for a garment of an avatar body that overlaps the avatar body in a finite volume.
31. The computing system of claim 27, wherein the modifying the second feature in the model includes projecting a three-dimensional garment model to form a two-dimensional garment view along a selected point of view to update the at least one pixel in the selected area based on the two-dimensional garment view.
32. The computing system of claim 27, wherein the modifying the first feature in the model includes verifying that at least one pixel of the at least one known garment collision is not included in the selected area when a selected point of view of the dressed avatar is modified.
33. A computer-implemented method, comprising: forming a projection of a dressed avatar in an immersive reality application running in a headset, wherein the dressed avatar includes a first 3D mesh for a body of the dressed avatar and at least one second 3D mesh for clothing of the dressed avatar; identifying, for the projection of the dressed avatar, an area that includes a garment collision; cropping a portion of the first 3D mesh, for the body of the dressed avatar, that includes the garment collision; and replacing the cropped portion of the first 3D mesh with part of the at least one second 3D mesh for clothing of the dressed avatar, to form a new projection of the dressed avatar.
34. The computer-implemented method of claim 33, wherein the forming the projection of the dressed avatar includes selecting a two-dimensional projection along a direction of view of a user of the headset.
35. The computer-implemented method of claim 33, wherein the identifying the area that includes the garment collision includes running a model trained to identify a garment collision on projections of dressed avatars.
36. The computer-implemented method of claim 33, wherein the identifying the area that includes the garment collision includes applying a 2D supervised pixel classification model based on a viewpoint of a user of the headset.
37. The computer-implemented method of claim 33, wherein the identifying the area that includes the garment collision includes predicting, for multiple pixels of the dressed avatar, a likelihood that the pixel indicates a body portion based on a pixel value and at least an adjacent pixel value.
38. The computer-implemented method of claim 33, wherein the replacing the cropped portion of the first 3D mesh includes applying coloring according to the at least one second 3D mesh.
39. The computer-implemented method of claim 33, wherein the identifying the area that includes the garment collision includes propagating a ray along a point of view of a user, in a direction of the area that includes the garment collision; and wherein the cropping the portion of the first 3D mesh includes removing a portion of first 3D mesh that intersects the ray.
40. The computer-implemented method of claim 33, wherein the identifying the area that includes the garment collision includes resolving an overlap between the first 3D mesh and the at least one second 3D mesh; and wherein the replacing the cropped portion includes projecting an updated 3D clothing model on the new projection of the dressed avatar.